



功率回路

内容: B+ 熔断器 / CHG、DSG 各 8 并联 NMOS / 栅极网络与 PNP 快速关断 / 分流器 / TVS / 预充 / P+ P- 端子

接口网络 (跨页用全局标签, 名称以本清单为准) :

- BAT+
- BAT-
- PACK+
- PACK-
- FET_MID
- CHG_G
- DSG_G
- SRP
- SRN
- NTC_FET
- NTC_SHUNT
- PRECHG_EN

状态: 待画。详细设计见 docs/03-详细设计.md ; 按 pcb-workflow 规则 00/00c/00d 执行。

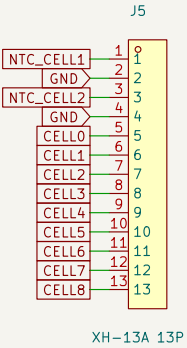
8S LiFeP04 200A BMS · MPS MP2797 + 开关电容均衡		
MP2645A-BMS		
Sheet: /功率回路/		
File: 01_power_path.kicad_sch		
Title: 功率回路		
Size: A3	Date: 2026-09-26	Rev: v0.3-skeleton
KiCad E.D.A. 10.0.6	Id: 2/6	

02 AFE:MP2797 采集与保护 (8S, C9-C16 并到 C8;WDT 悬空;xALERT 高有效 100k 下拉)

U1 = MP2797DFP-0000-T (C18198873, TQFP-48)。本页 U1 全部引脚以「短线 + 标签」引出;未用脚放 NC 标志 (29 WDT / 33 SDO / 34-37 GPIO / 40 N/C / 41 SBYDSG)。
C9-C16 (脚 5,4,3,2,1,48,47,46) 用一条竖线并在 C8 (脚 6) 的 VC8 上, 不串电阻、不加电容 (Table 23「directly connect」)。
AFE_3V3 只接 C215, 不与板上 +3V3 相连;NTC 四路上拉 (R216-R219) 都接 NTCB, 不接任何 3V3。

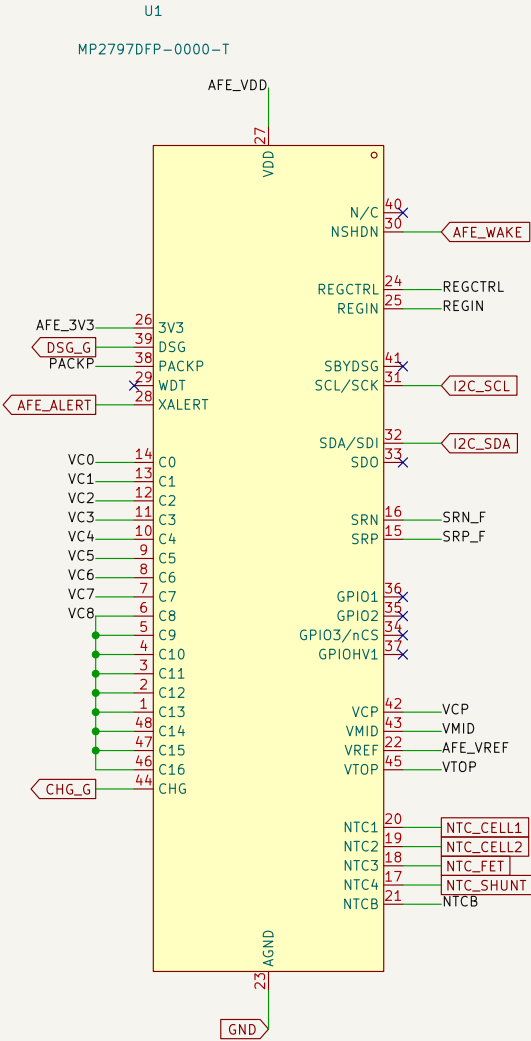
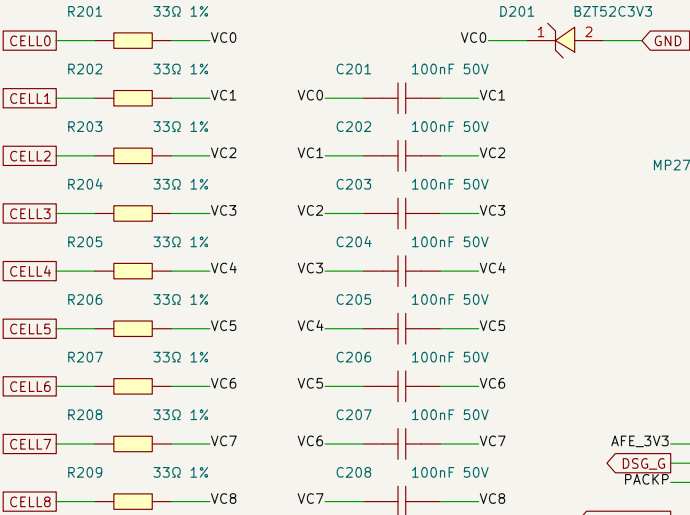
§10 J5 采样端子

C40669 XH 2.50mm 13P 立式;
1/3 = NTC_CELL1/2, 2/4 = GND;
5-13 = CELLO,..,CELL8;
NTC 放低压端, 线束短路
不会把 29V 灌进 NTC 脚。

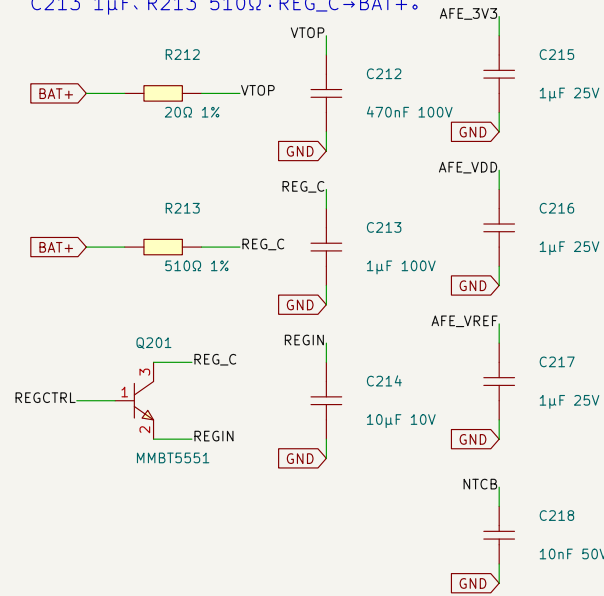


§4 电芯采样 (8 串 9 路)

J5.5-13 CELLO-CELL8 → R201-R209 33Ω 1% → VC0-VC8 → U1.14/13/12/11/10/9/8/7/6。
C201-C208 100nF/50V 跨相邻两个 VC (8 只, 不接地); C0 只挂 D201 稳压 (阴极→VC0、阳极→GND)。
均衡电流走串阻: 33Ω 时典型 39mA、最坏 45mA, 0805 耗散余量 1.5 倍; 本页不出现任何 BAL* 标签。

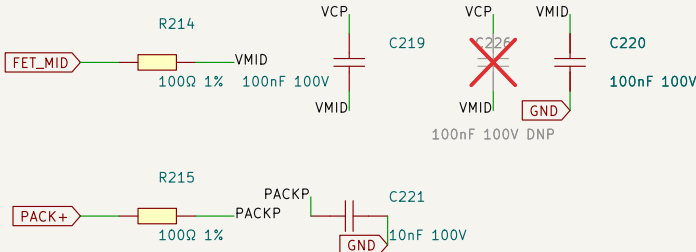


§9 电源: BAT+ 供电 (U-1)
U1.45 VTOP ← R212 20Ω ← BAT+; C212 470nF/100V 到 GND。
U1.25 REGIN ← Q201(MMBT5551) 发射极 (C214 10μF/10V);
Q201 基极 = REGCTRL, 集电极 = REG_C;
C213 1μF, R213 510Ω; REG_C→BAT+。



§6 高边驱动 (CHG/DSG/VCP/VMID/PACKP)

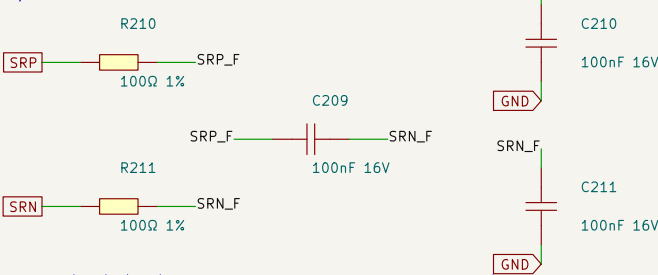
U1.43 VMID ← R214 100Ω ← FET_MID, C220 100nF/100V 到 GND。
U1.38 PACKP ← R215 100Ω ← PACK+, C221 10nF/100V 到 GND (Table 18「不得增减」)。



U1.42 VCP ↔ VMID 跨接 C219 100nF/100V;
C226 为扩展焊盘 (DNP, 样板实测 VCP 跌落后再定)。
U1.44 CHG、U1.39 DSG 本页不申件,
R_DRV/DZ/PNP 快关在 01 页。

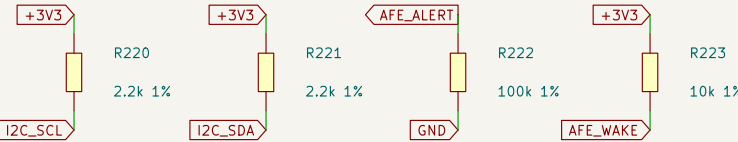
§5 电流采样 (SRP/SRN 开尔文)

SRP/SRN 取自分流器电池侧 (01 页),
经 R210/R211 100Ω → SRP_F/SRN_F → U1.15/U1.16;
C209 100nF 跨 SRP_F→SRN_F (差模);
C210/C211 各 100nF 到 GND (共模)。



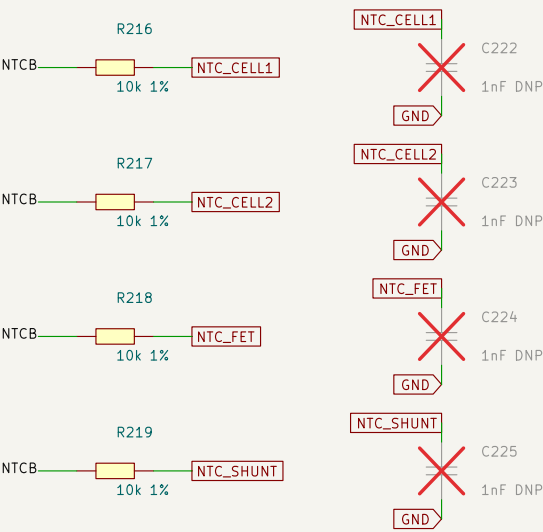
§8 通讯与中断 (I2C / xALERT / NSHDN)

R220/R221 = 2.2k 1% 上拉到 +3V3 (I2C 开漏必需);
R222 = 100k 1% 把 AFE_ALERT 下拉到 GND (xALERT 推挽、
ALERT_POL 默认高有效, 空闲低电平);
R223 = 10k 1% 把 AFE_WAKE 上拉到 +3V3 (不是 AFE_3V3)。
WDT (U1.29) 悬空, 已在 U1 侧放 NC 标志。



§7 NTC ×4 (电芯 2 路 + FET / 分流器)

R216-R219 = 10k 1% 上拉到 NTCB (不是 3V3); C222-C225 = 1nF 为 DNP。
(NTCB_DYNAMIC_ON=1; 装了会让低温读数偏低)
NTC_CELL1/2 走 J5.1/2 与 J5.3/4 的线束探头;
NTC_FET → 01 页 RT3; NTC_SHUNT → 01 页 RT4。



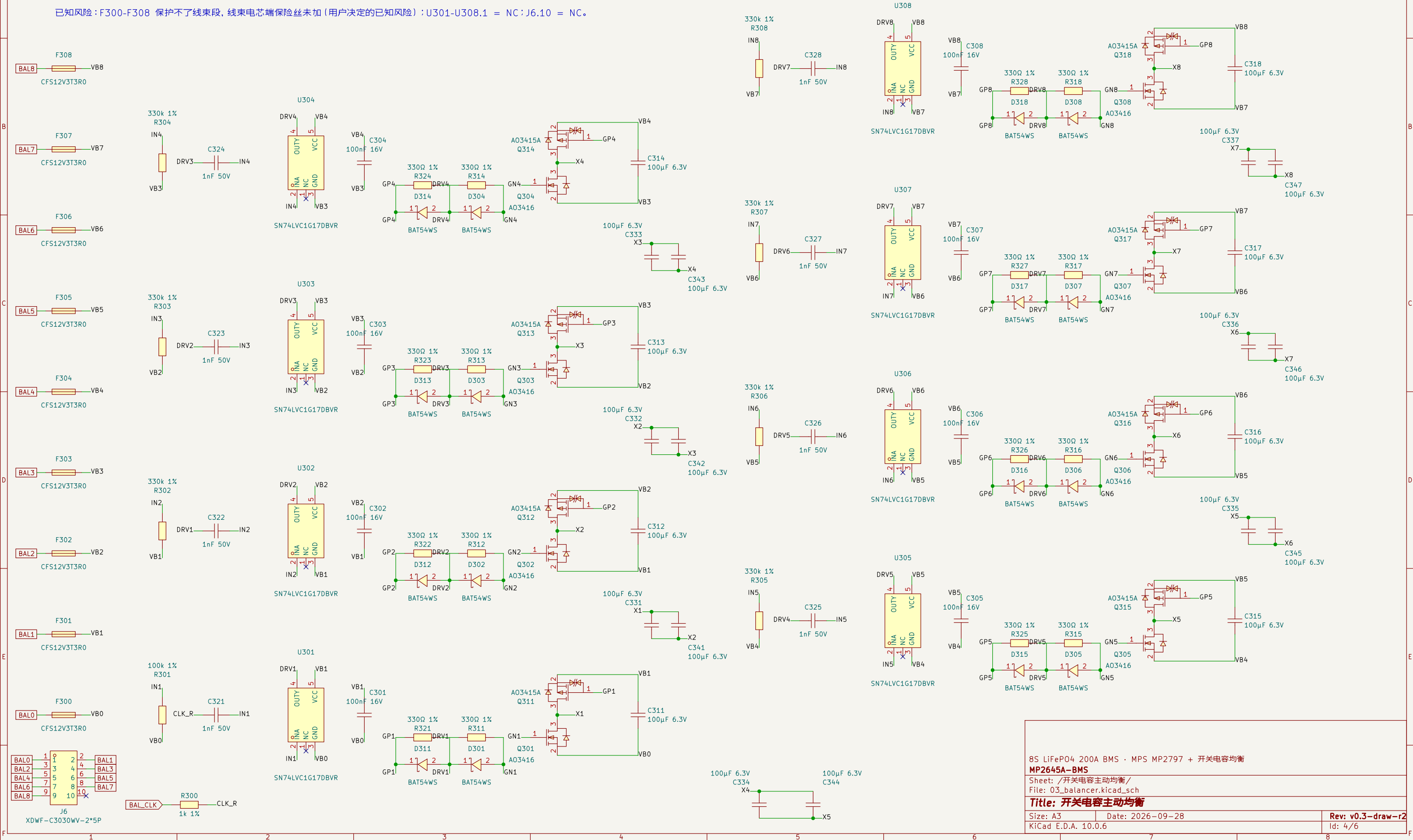
开关电容主动均衡（8 级半桥 + 飞电容, 75kHz）

内容：8 个互补半桥 QN/QP (AO3416/AO3415A) + 8 个 74LVC1G17 (本节电芯供电) + 7 级飞电容 2×100μF；每节 C31k 100μF 本地储能。
全级交流耦合：BAL_CLK 经 1nF (C321-C328) 逐级耦合，首级亦然 (VB0 不接地)；均衡节点经 3A 熔丝到 BAL0-BAL8；本页无 GND、无电源符号。
死区：R31k/R32k 330Ω ∥ D30k/D31k BAT54WS (慢开快关, CT≤10pF) 防上下管直通；停时钟/复位：R301 100k、R302-R308 330k 下拉，第 1 级先泄放完避免混相。

接口网络（跨页用全局标签, 名称以本清单为准）：

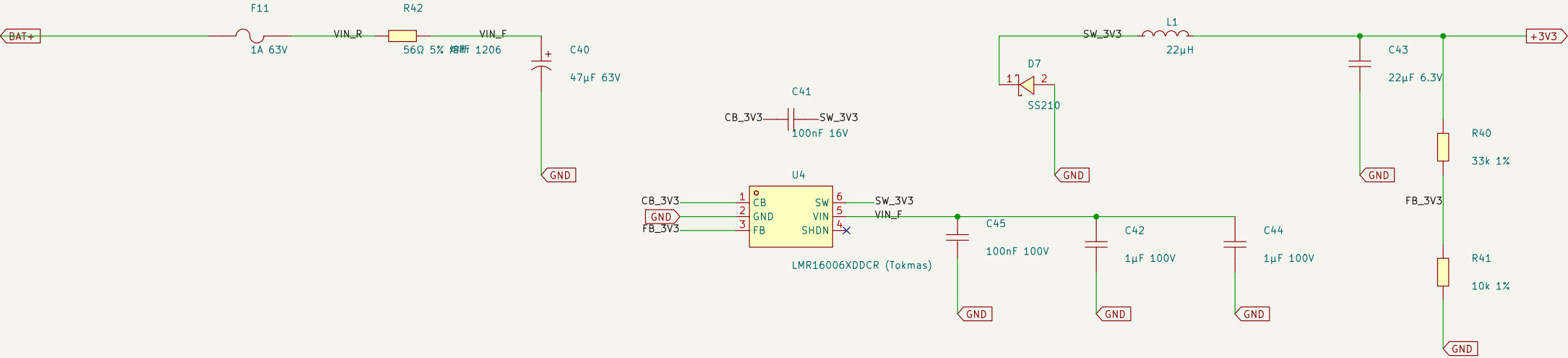
· BAL0..BAL8（均衡节点, 经 F300-F308）；· BAL_CLK（时钟输入）

已知风险：F300-F308 保护不了线束段，线束电芯端保险丝未加（用户决定的已知风险）：U301-U308.1 = NC；J6.10 = NC。



辅助电源 (LMR16006X 直降 3.3V)

LMR16006X 600kHz (Tokmas) 非同步 buck; D7 续流必需; SHDN 悬空=常开; 输入 C40 47μF/63V + C42/C44 2×1μF/100V
R42 56Ω 熔断电阻与 C40 构成输入 RC; 限上电浪涌与短路电流, VIN_F 短路时受控开路; C45 100nF/100V 贴 VIN (Tokmas 手册 p14)



MCU 与通讯 (CH32V203C8T6 + CAN)

CH32V203C8T6 144MHz (12MHz×12) ;CAN1 PA11/12→SN65HVD230 (3.3V) ;I2C1 PB6/7→02 页 MP2797 ;

BAL_CLK=PA8(TIM1_CH1 75kHz)+R52 100k 下拉;PRECHG_EN=PA1+R51 100k 下拉;AFE_WAKE=MP2797 NSHDN(PB5 开漏, 上拉 R_WAK 在 02 页);

差分对 120Ω (规则 0c : J7→U8→L2/R56/R57→U7 直通, 顶层不换层) ;JP1 插帽接入 120Ω 终端, 拔帽不用;J8 DNP, 下载器勿从 1 脚供电。

I2C 上拉 R_I2C1/2 在 02 页。

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I2C 上拉 R_I2C1/2 在 02 页。

8S LiFePO4 200A BMS · MPS MP2797 + 开关电源均衡

MP2645A-BMS

Sheet: /MCU 与通讯/

File: 05_mcu_comm.kicad_sch

Title: MCU 与通讯

Size: A3

Date: 2026-09-27

Rev: v0.3-draw-r1

KiCad E.D.A. 10.0.6

Id: 6/6

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I2C 上拉 R_I2C1/2 在 02 页。

The schematic diagram illustrates the MCU and communication circuit for the CH32V203C8T6. The central component is the MCU (U6), which is connected to various external components. The MCU pins are labeled with their functions and connections to the external components. The diagram includes a 12MHz oscillator (Y1) connected to the HSE_IN pin. The HSE_OUT pin is connected to a 33pF capacitor (C57). The PRECHG_EN pin is connected to a 100k resistor (R51) and a 100nF capacitor (C50). The BAL_CLK pin is connected to a 100k resistor (R52) and a 100nF capacitor (C51). The VBAT pin is connected to a 100nF capacitor (C52). The HSE_IN pin is connected to a 100nF capacitor (C53). The VDDA pin is connected to a 100nF capacitor (C54). The PA0/WKUP/ADC0 pin is connected to a 100nF capacitor (C55). The PA1/ADC1 pin is connected to a 100nF capacitor (C56). The PA2/ADC2 pin is connected to a 100nF capacitor (C57). The PA3/ADC3 pin is connected to a 100nF capacitor (C58). The PA4/ADC4/DAC0 pin is connected to a 100nF capacitor (C59). The PA5/ADC5/DAC1 pin is connected to a 100nF capacitor (C60). The PA6/ADC6 pin is connected to a 100nF capacitor (C61). The PA7/ADC7 pin is connected to a 100nF capacitor (C62). The PA8/ADC8 pin is connected to a 100nF capacitor (C63). The PA9/ADC9 pin is connected to a 100nF capacitor (C64). The PA10/ADC10 pin is connected to a 100nF capacitor (C65). The PA11/ADC11 pin is connected to a 100nF capacitor (C66). The PA12/ADC12 pin is connected to a 100nF capacitor (C67). The PA13/ADC13 pin is connected to a 100nF capacitor (C68). The PA14/ADC14 pin is connected to a 100nF capacitor (C69). The PA15/ADC15 pin is connected to a 100nF capacitor (C70). The PA16/ADC16 pin is connected to a 100nF capacitor (C71). The PA17/ADC17 pin is connected to a 100nF capacitor (C72). The PA18/ADC18 pin is connected to a 100nF capacitor (C73). The PA19/ADC19 pin is connected to a 100nF capacitor (C74). The PA20/ADC20 pin is connected to a 100nF capacitor (C75). The PA21/ADC21 pin is connected to a 100nF capacitor (C76). The PA22/ADC22 pin is connected to a 100nF capacitor (C77). The PA23/ADC23 pin is connected to a 100nF capacitor (C78). The PA24/ADC24 pin is connected to a 100nF capacitor (C79). The PA25/ADC25 pin is connected to a 100nF capacitor (C80). The PA26/ADC26 pin is connected to a 100nF capacitor (C81). The PA27/ADC27 pin is connected to a 100nF capacitor (C82). The PA28/ADC28 pin is connected to a 100nF capacitor (C83). The PA29/ADC29 pin is connected to a 100nF capacitor (C84). The PA30/ADC30 pin is connected to a 100nF capacitor (C85). The PA31/ADC31 pin is connected to a 100nF capacitor (C86). The PA32/ADC32 pin is connected to a 100nF capacitor (C87). The PA33/ADC33 pin is connected to a 100nF capacitor (C88). The PA34/ADC34 pin is connected to a 100nF capacitor (C89). The PA35/ADC35 pin is connected to a 100nF capacitor (C90). The PA36/ADC36 pin is connected to a 100nF capacitor (C91). The PA37/ADC37 pin is connected to a 100nF capacitor (C92). The PA38/ADC38 pin is connected to a 100nF capacitor (C93). The PA39/ADC39 pin is connected to a 100nF capacitor (C94). The PA40/ADC40 pin is connected to a 100nF capacitor (C95). The PA41/ADC41 pin is connected to a 100nF capacitor (C96). The PA42/ADC42 pin is connected to a 100nF capacitor (C97). The PA43/ADC43 pin is connected to a 100nF capacitor (C98). The PA44/ADC44 pin is connected to a 100nF capacitor (C99). The PA45/ADC45 pin is connected to a 100nF capacitor (C100). The PA46/ADC46 pin is connected to a 100nF capacitor (C101). The PA47/ADC47 pin is connected to a 100nF capacitor (C102). The PA48/ADC48 pin is connected to a 100nF capacitor (C103). The PA49/ADC49 pin is connected to a 100nF capacitor (C104). The PA50/ADC50 pin is connected to a 100nF capacitor (C105). The PA51/ADC51 pin is connected to a 100nF capacitor (C106). 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The PA68/ADC68 pin is connected to a 100nF capacitor (C123). The PA69/ADC69 pin is connected to a 100nF capacitor (C124). The PA70/ADC70 pin is connected to a 100nF capacitor (C125). The PA71/ADC71 pin is connected to a 100nF capacitor (C126). The PA72/ADC72 pin is connected to a 100nF capacitor (C127). The PA73/ADC73 pin is connected to a 100nF capacitor (C128). The PA74/ADC74 pin is connected to a 100nF capacitor (C129). The PA75/ADC75 pin is connected to a 100nF capacitor (C130). The PA76/ADC76 pin is connected to a 100nF capacitor (C131). The PA77/ADC77 pin is connected to a 100nF capacitor (C132). The PA78/ADC78 pin is connected to a 100nF capacitor (C133). The PA79/ADC79 pin is connected to a 100nF capacitor (C134). The PA80/ADC80 pin is connected to a 100nF capacitor (C135). The PA81/ADC81 pin is connected to a 100nF capacitor (C136). The PA82/ADC82 pin is connected to a 100nF capacitor (C137). The PA83/ADC83 pin is connected to a 100nF capacitor (C138). The PA84/ADC84 pin is connected to a 100nF capacitor (C139). The PA85/ADC85 pin is connected to a 100nF capacitor (C140). The PA86/ADC86 pin is connected to a 100nF capacitor (C141). The PA87/ADC87 pin is connected to a 100nF capacitor (C142). The PA88/ADC88 pin is connected to a 100nF capacitor (C143). The PA89/ADC89 pin is connected to a 100nF capacitor (C144). The PA90/ADC90 pin is connected to a 100nF capacitor (C145). The PA91/ADC91 pin is connected to a 100nF capacitor (C146). The PA92/ADC92 pin is connected to a 100nF capacitor (C147). The PA93/ADC93 pin is connected to a 100nF capacitor (C148). The PA94/ADC94 pin is connected to a 100nF capacitor (C149). The PA95/ADC95 pin is connected to a 100nF capacitor (C150). The PA96/ADC96 pin is connected to a 100nF capacitor (C151). The PA97/ADC97 pin is connected to a 100nF capacitor (C152). The PA98/ADC98 pin is connected to a 100nF capacitor (C153). The PA99/ADC99 pin is connected to a 100nF capacitor (C154). The PA100/ADC100 pin is connected to a 100nF capacitor (C155). The PA101/ADC101 pin is connected to a 100nF capacitor (C156). The PA102/ADC102 pin is connected to a 100nF capacitor (C157). The PA103/ADC103 pin is connected to a 100nF capacitor (C158). The PA104/ADC104 pin is connected to a 100nF capacitor (C159). The PA105/ADC105 pin is connected to a 100nF capacitor (C160). The PA106/ADC106 pin is connected to a 100nF capacitor (C161). The PA107/ADC107 pin is connected to a 100nF capacitor (C162). The PA108/ADC108 pin is connected to a 100nF capacitor (C163). The PA109/ADC109 pin is connected to a 100nF capacitor (C164). The PA110/ADC110 pin is connected to a 100nF capacitor (C165). The PA111/ADC111 pin is connected to a 100nF capacitor (C166). The PA112/ADC112 pin is connected to a 100nF capacitor (C167). The PA113/ADC113 pin is connected to a 100nF capacitor (C168). The PA114/ADC114 pin is connected to a 100nF capacitor (C169). The PA115/ADC115 pin is connected to a 100nF capacitor (C170). The PA116/ADC116 pin is connected to a 100nF capacitor (C171). The PA117/ADC117 pin is connected to a 100nF capacitor (C172). The PA118/ADC118 pin is connected to a 100nF capacitor (C173). The PA119/ADC119 pin is connected to a 100nF capacitor (C174). The PA120/ADC120 pin is connected to a 100nF capacitor (C175). The PA121/ADC121 pin is connected to a 100nF capacitor (C176). The PA122/ADC122 pin is connected to a 100nF capacitor (C177). The PA123/ADC123 pin is connected to a 100nF capacitor (C178). The PA124/ADC124 pin is connected to a 100nF capacitor (C179). The PA125/ADC125 pin is connected to a 100nF capacitor (C180). The PA126/ADC126 pin is connected to a 100nF capacitor

[illegible]

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